

OPTIMOLOL® EYE DROPS 0.5% w/v

Ingredient(s):

Each ml contains:

Timolol Maleate	6.84mg
(eq. to Timolol 5mg)	
Benzalkonium Chloride (as preservative)	0.05mg

Pharmacodynamics:

Mechanism of action

Timolol maleate is a non-selective beta-adrenergic receptor blocking agent that does not have significant intrinsic sympathomimetic, direct myocardial depressant, or local anaesthetic activity. Timolol maleate combines reversibly with the beta-adrenergic receptor, and this inhibits the usual biologic response that would occur with stimulation of that receptor. This specific competitive antagonism blocks stimulation of the beta-adrenergic stimulating (agonist) activity, whether these originate from an endogenous or exogenous source. Reversal of this blockade can be accomplished by increasing the concentration of the agonist which will restore the usual biological response.

Clinical efficacy and safety:

Unlike miotics, OPTIMOLOL reduces intraocular pressure (IOP) with little or no effect on accommodation or pupil size. In patients with cataracts, the inability to see around lenticular opacities when the pupil is constricted is avoided. When changing patients from miotics to OPTIMOLOL a refraction might be necessary when the effects of the miotic have passed. Diminished response after prolonged therapy with OPTIMOLOL has been reported in some patients.

Paediatric Population:

There is only very limited data available on the use of timolol (0.25%, 0.5% twice daily one drop) in the paediatric population. However, available data have shown to some extent evidence, that timolol in the indication primary congenital and primary juvenile glaucoma is effective in short term treatment.

Pharmacokinetics:

The onset of reduction in intraocular pressure can be detected within one-half hour after a single dose. The maximum effect occurs in one or two hours; significant lowering of IOP can be maintained for as long as 24 hours with a single dose.

Paediatric Population:

As already confirmed by adult data, 80% of each eye drop passes through the nasolacrimal system where it may be rapidly absorbed into the systemic circulation via the nasal mucosa, conjunctiva, nasolacrimal duct, oropharynx and gut, or the skin from tear overflow.

Due to the fact that the blood volume in children is smaller than that in adults a higher circulation concentration has to be taken into account. In addition, neonates have immature metabolic enzyme pathways and it may result in an increase in elimination half-life and potentiating adverse events.

Limited data show that plasma timolol levels in children after 0.25% greatly exceed those in adults after 0.5%, especially in infants and are presumed to increase the risk of side effects such as bronchospasm and bradycardia.

Indication:

OPTIMOLOL Eye Drops 0.5% w/v is indicated in the treatment of elevated intraocular pressure in patients with ocular hypertension or open angle glaucoma.

Recommended Dose:

Adults and children over 12 years

Recommended therapy is one drop of OPTIMOLOL in the affected eye(s) twice a day.

OPTIMOLOL is available in 0.5% strength. It is not able to deliver other recommended dose of Timolol.

Please source alternative product for other recommended dosage.

Elderly

Dosage need not be modified for the elderly as there has been wide experience with the use of OPTIMOLOL in elderly patients. When using nasolacrimal occlusion or closing the eyelids for 2 minutes, the systemic absorption is reduced. This may result in a decrease in systemic side effects and an increase in local activity. Intraocular pressure should be reassessed approximately four weeks after starting treatment because response to OPTIMOLOL may take a few weeks to stabilise. Provided that intraocular pressure is maintained at satisfactory levels, many patients can then be placed on once daily therapy. If necessary, concomitant treatment with miotics, epinephrine and/or carbonic anhydrase inhibitors can be instituted. In order to prevent the active substance(s) from being washed out when additional ophthalmic medication is used, an interval of at least 10 minutes between each application is recommended. The use of two topical beta-adrenergic agents is not recommended. Transfer from other topical beta-blocking agents: Discontinue use after a full day of therapy and start treatment with OPTIMOLOL the next day, with one drop in each affected eye twice daily. Transfer from a single antiglaucoma agent other than a topical beta-blocking agent: Continue the agent and add one drop of OPTIMOLOL in each affected eye twice daily. On the following day, discontinue the previous agent completely, and continue with OPTIMOLOL. Patients should be instructed to remove soft contact lenses before using timolol.

Paediatric Population

Due to limited data, Timolol could only be recommended for use in Primary congenital and primary juvenile glaucoma for a transitional period while decision is made on a surgical approach and in case of failed surgery while awaiting further options.

Posology

Clinicians should strongly evaluate the risks and benefits when considering medical therapy with Timolol in paediatric patients. A detailed paediatric history and examination to determine the presence of systemic abnormalities should precede the use of Timolol. No specific dosage recommendation can be given as there is only limited clinical data. However, if benefit outweighs the risk, it is recommended to use the lowest active agent concentration available once daily. If IOP could not be sufficiently controlled, a careful up titration to a maximum of two drops daily per affected eye has to be considered. If applied twice daily, an interval of 12 hours should be preferred. Furthermore the patients, especially neonates, should be strongly observed after the first dose for one to two hours in the office and closely monitored for ocular and systemic side effects until surgery is performed. With regard to pediatrics use, the 0.1% active agent concentration might already be sufficient.

Method of administration

To limit potential adverse effects only one drop should be instilled per dosing time. Systemic absorption of topically administered beta-blockers can be reduced by nasolacrimal occlusion and by keeping the eyes closed as long as possible after instillation of drops.

Duration of treatment

For a transient treatment in the paediatric population.

Usage Instruction

1. Ensure that your hands are clean.
2. To open the bottle, unscrew the cap of the bottle.

3. Check the tip to make sure it is not cracked or chipped.
4. Tilt your head back and pull your lower eyelid down with your finger to form a pocket.
5. Invert the container, press lightly until a single drop is dispensed into the eye as directed by your doctor. **DO NOT TOUCH YOUR EYE OR EYELID WITH THE DROPPER TIP.**
6. Repeat steps 4 and 5 with the other eye if instructed to do so by your doctor.
7. Replace the cap by turning until it is firmly touching the bottle.

Route of Administration

For Ophthalmic Use Only.

Contraindication:

OPTIMOLOL is contraindicated in patients with:

- Hypersensitivity to the active substance, or to any of the excipients listed.
- Reactive airway disease including bronchial asthma or a history of bronchial asthma, severe chronic obstructive pulmonary disease.
- Sinus bradycardia, sick sinus syndrome sino-atrial block, second or third degree atrioventricular block not controlled with pace-maker.
- Overt cardiac failure, cardiogenic shock.

Warnings and Precautions:

Like other topically applied ophthalmic agents, timolol is absorbed systemically. Due to beta-adrenergic component, timolol, the same types of cardiovascular, pulmonary and other adverse reactions seen with systemic beta-adrenergic blocking agents may occur. Incidence of systemic adverse drug reactions (ADRs) after topical ophthalmic administration is lower than for systemic administration.

Cardiac disorders

In patients with cardiovascular diseases (e.g. coronary heart disease, Prinzmetal's angina and cardiac failure) and hypotension therapy with beta-blockers should be critically assessed and the therapy with other active substances should be considered. Patients with cardiovascular diseases should be watched for signs of deterioration of these diseases and of adverse reactions.

Due to its negative effect on conduction time, beta-blockers should only be given with caution to patients with first degree heart block. Cardiac failure should be adequately controlled before beginning therapy with OPTIMOLOL. Patients with a history of severe cardiac disease should be watched for signs of cardiac failure and have their pulse rates monitored.

Vascular disorders

Patients with severe peripheral circulatory disturbance/disorders (i.e. severe forms of Raynaud's disease or Raynaud's syndrome) should be treated with caution.

Respiratory disorders

Respiratory reactions, including death due to bronchospasm in patients with asthma have been reported following administration of some ophthalmic beta-blockers.

OPTIMOLOL should be used with caution, in patients with mild/moderate chronic obstructive pulmonary disease (COPD) and only if the potential benefit outweighs the potential risk.

Hypoglycaemia/diabetes

Beta-blockers should be administered with caution in patients subject to spontaneous hypoglycaemia or to patients with labile diabetes, as beta-blockers may mask the signs and symptoms of acute hypoglycaemia.

Beta-blockers may also mask the signs of hyperthyroidism.

Corneal diseases

Ophthalmic beta-blockers may induce dryness of eyes. Patients with corneal diseases should be treated with caution.

Other beta-blocking agents

The effect on intraocular pressure or the known effects of systemic beta-blockade may be potentiated when timolol is given to the patients already receiving a systemic beta-blocking agent. The response of these patients should be closely observed. The use of two topical beta-adrenergic blocking agents is not recommended.

There have been reports of skin rashes and/or dry eyes associated with the use of beta-adrenoreceptor blocking drugs. The reported incidence is small and in most cases the symptoms have cleared when treatment was withdrawn. Discontinuation of the drug should be considered if any such reaction is not otherwise explicable. Cessation of therapy involving beta-blockade should be gradual.

Anaphylactic reactions

While taking beta-blockers, patients with history of atopy or a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge with such allergens and unresponsive to the usual dose of epinephrine (adrenaline) used to treat anaphylactic reactions.

Choroidal detachment

Choroidal detachment has been reported with administration of aqueous suppressant therapy (e.g. timolol, acetazolamide) after filtration procedures.

Surgical anaesthesia

Beta-blocking ophthalmological preparations may block systemic beta-agonist effects e.g. of epinephrine (adrenaline). The anaesthesiologist should be informed when the patient is receiving timolol.

OPTIMOLOL has been generally well tolerated in glaucoma patients wearing conventional hard contact lenses. OPTIMOLOL has not been studied in patients wearing lenses made with material other than polymethylmethacrylate (PMMA), which is used to make hard contact lenses.

OPTIMOLOL contains benzalkonium chloride as a preservative which may be deposited in soft contact lenses; therefore OPTIMOLOL should not be used while wearing these lenses. The lenses should be removed before application of the drops and not reinserted earlier than 15 minutes after use.

Benzalkonium chloride has been reported to cause eye irritation, symptoms of dry eyes and may affect the tear film and corneal surface. Should be used with caution in dry eye patients and in patients where the cornea may be compromised. Patients should be monitored in case of prolonged use.

In patients with angle-closure glaucoma, the immediate objective of treatment is to reopen the angle. This requires constricting the pupil with a miotic. OPTIMOLOL has little or no effect on the pupil. When OPTIMOLOL is used to reduce elevated intraocular pressure in angle-closure glaucoma it should be used with a miotic and not alone.

A reduction in ocular hypotensive response has been reported in some patients following prolonged therapy with Timolol maleate eye drops.

Muscle weakness: Beta-adrenergic blockade has been reported to potentiate muscle weakness consistent with certain myasthenic symptoms (e.g. diplopia, ptosis, and generalised weakness).

Timolol have been reported rarely to increase muscle weakness in some patients with myasthenia gravis or myasthenic symptoms.

Patients should be instructed to avoid allowing the tip of the dispensing container to contact the eye or surrounding structures.

Patients should also be instructed that ocular solutions, if handled improperly can become contaminated by common bacteria known to cause ocular infections. Serious damage to the eye and subsequent loss of vision may result from using contaminated solutions.

Patients should be advised that if they develop an intercurrent ocular condition (e.g. trauma, ocular surgery or infection), they should immediately seek their physician's advice concerning the continued use of the present multidose container.

There have been reports of bacterial keratitis associated with the use of multiple dose containers of topical ophthalmic products. These containers had been inadvertently contaminated by patients who, in most cases, had a concurrent corneal disease or a disruption of the ocular epithelial surface.

Paediatric Population

Timolol solutions should generally be used cautiously in young glaucoma patients.

It is important to notify the parents of potential side effects so they can immediately discontinue the drug therapy. Signs to look for are, for example, coughing and wheezing.

Because of the possibility of apnoea and Cheyne-Stokes breathing, the drug should be used with extreme caution in neonates, infants and younger children. A portable apnoea monitor may also be helpful for neonates on timolol.

Interactions with Other Medicaments:

No specific drug interaction studies have been performed with timolol maleate.

There is a potential for additive effects resulting in hypotension and/or marked bradycardia when ophthalmic beta-blockers solution is administered concomitantly with oral calcium-channel blockers, beta-adrenergic blocking agents, antiarrhythmics (including amiodarone), digitalis glycosides, rauwolfia alkaloids, parasympathomimetics, guanethidine.

Although OPTIMOLOL alone has little or no effect on pupil size, mydriasis resulting from concomitant use of ophthalmic beta-blockers and epinephrine (adrenaline) has been reported occasionally.

Potentiated systemic beta-blockade (e.g. decreased heart rate, depression) has been reported during combined treatment with CYP2D6 inhibitors (e.g. quinidine, fluoxetine, paroxetine) and timolol.

Oral beta-adrenergic blocking agents may exacerbate the rebound hypertension which can follow the withdrawal of clonidine.

Close observation of the patient is recommended when a beta-blocker is administered to patients receiving catecholamine-depleting drugs such as reserpine, because of possible additive effects and the production of hypotension and/or marked bradycardia, which may produce vertigo, syncope, or postural hypotension.

Oral calcium-channel antagonists may be used in combination with beta-adrenergic blocking agents when heart function is normal, but should be avoided in patients with impaired cardiac function.

The potential exists for hypotension, AV conduction disturbances and left ventricular failure to occur in patients receiving a beta-blocking agent when an oral calcium-channel blocker is added to the treatment regimen. The nature of any cardiovascular adverse effects tends to depend on the type of calcium-channel blocker used. Dihydropyridine derivatives, such as nifedipine, may lead to hypotension, whereas verapamil or diltiazem have a greater propensity to lead to AV conduction disturbances or left ventricular failure when used with a beta-blocker.

Intravenous calcium channel blockers should be used with caution in patients receiving beta-adrenergic blocking agents.

The concomitant use of beta-adrenergic blocking agents and digitalis with either diltiazem or verapamil may have additive effects in prolonging AV conduction time.

Beta-blockers may increase the hypoglycaemic effect of anti-diabetic agents. Beta-blockers can mask the signs and symptoms of hypoglycaemia.

Pregnancy, Lactation and Fertility

Pregnancy

There are no adequate data for the use of timolol maleate in pregnant women. OPTIMOLOL should not be used during pregnancy unless clearly necessary.

Epidemiological studies have not revealed malformative effects but show a risk for intra uterine growth retardation when beta-blockers are administered by the oral route. In addition, signs and symptoms of beta-blockade (e.g. bradycardia, hypotension, respiratory distress and hypoglycaemia) have been observed in the neonate when beta-blockers have been administered until delivery. If OPTIMOLOL is administered until delivery, the neonate should be carefully monitored during the first days of life.

Lactation

Timolol is detectable in human milk. A decision for breastfeeding mothers, either to stop taking OPTIMOLOL or stop nursing, should be based on the importance of the drug to the mother.

Fertility

Timolol have not been found to have any effect on male or female fertility in animal studies.

Undesirable Effects:

Like other topically applied ophthalmic drugs, timolol maleate is absorbed into the systemic circulation. This may cause similar undesirable effects as seen with systemic beta-blocking agents. Incidence of systemic ADRs after topical ophthalmic administration is lower than for systemic administration. The following adverse reactions have been reported with ocular administration of timolol maleate formulations, either in clinical trials or since the drug has been marketed.

Additional side effects have been reported in clinical experiences with systemic timolol maleate, and may be considered potential effects of ophthalmic timolol maleate. Also listed are adverse reactions seen within the class of ophthalmic beta-blockers and may potentially occur with timolol.

Eye disorders

ocular: signs and symptoms of ocular irritation, (e.g. burning, stinging, itching, tearing, redness), conjunctivitis, blepharitis, keratitis, dry eyes, decreased corneal sensitivity, blurred vision, corneal erosion. Visual disturbances, including refractive changes (due to withdrawal of miotic therapy in some cases), diplopia, ptosis and choroidal detachment following filtration surgery. Cases of corneal calcification have been reported very rarely in association with the use of phosphate containing eye drops in some patients with significantly damaged corneas.

Ear and labyrinth disorders

ocular: tinnitus

Cardiac disorders

ocular: bradycardia, chest pain, arrhythmia, heart block, congestive heart failure, palpitations, cardiac arrest, cardiac failure, oedema.

systemic: atrioventricular block (second- or third-degree), sino-atrial block, pulmonary oedema, worsening of arterial insufficiency, worsening of angina pectoris, vasodilation.

Vascular disorders:

ocular: claudication, hypotension, Raynaud's phenomenon, cold hands and feet.

Respiratory, thoracic and mediastinal disorders:

ocular: bronchospasm (predominantly in patients with pre-existing bronchospastic disease), respiratory failure, dyspnoea, cough;
systemic: rales.

General disorders and administration site conditions:

ocular: asthenia, fatigue;
systemic: extremity pain, decreased exercise tolerance.

Skin and subcutaneous tissue disorders

ocular: alopecia, psoriasiform rash or exacerbation of psoriasis, skin rash;
systemic: sweating, exfoliative dermatitis.

Immune system disorders:

ocular: systemic lupus erythematosus, pruritus;
systemic: signs and symptoms of allergic reactions including anaphylaxis, angioedema, urticaria, localised and generalised rash, anaphylactic reaction.

Psychiatric disorders

ocular: depression, insomnia, nightmares, memory loss, hallucination;
systemic: diminished concentration, increased dreaming.

Nervous system disorders

ocular: syncope, cerebrovascular accident, cerebral ischemia, headache, dizziness, increase in signs and symptoms of myasthenia gravis, paraesthesia;
systemic: vertigo, local weakness

Gastrointestinal disorders:

ocular: nausea, diarrhoea, dyspepsia, dry mouth, dysgeusia, abdominal pain, vomiting.

Reproductive system and breast disorders:

ocular: decreased libido, Peyronie's disease, sexual dysfunction such as impotence;
systemic: micturition difficulties.

Metabolism and nutrition disorders:

ocular: hypoglycaemia;
systemic: hyperglycaemia.

Musculoskeletal and connective tissue disorders:

ocular: myalgia;
systemic: arthralgia.

Blood and lymphatic system disorders:

systemic: non-thrombocytopenic purpura.

Effects on Ability to Drive and Use Machine

No studies on the effect of this medicinal product on the ability to drive have been conducted. While driving vehicles or operating different machines, it should be taken into account that occasionally visual disturbances may occur including refractive changes, diplopia, ptosis, frequent episodes of mild and transient blurred vision and occasional episodes of dizziness or fatigue.

Overdose and Treatment

There have been reports of inadvertent overdosage with OPTIMOLOL resulting in systemic effects similar to those seen with systemic beta-adrenergic blocking agents such as dizziness, headache, shortness of breath, bradycardia, bronchospasm, and cardiac arrest.

Management

If overdosage occurs, the following measures should be considered:

1. Gastric lavage, if ingested. Studies have shown that timolol does not dialyse readily.
2. Symptomatic bradycardia: Atropine sulphate, 0.25 to 2mg intravenously, should be used to induce vagal blockade. If bradycardia persists, intravenous isoprenaline hydrochloride should be administered cautiously. In refractory cases, the use of a cardiac pacemaker may be considered.
3. Hypotension: A sympathomimetic pressor agent such as dopamine, dobutamine or noradrenaline should be used. In refractory cases, the use of glucagon has been reported to be useful.
4. Bronchospasm: Isoprenaline hydrochloride should be used. Additional therapy with aminophylline may be considered.
5. Acute cardiac failure: Conventional therapy with digitalis, diuretics and oxygen should be instituted immediately. In refractory cases, the use of intravenous aminophylline is suggested. This may be followed, if necessary, by glucagon which has been reported to be useful.
6. Heart block (second or third degree): Isoprenaline hydrochloride or a pacemaker should be used.

Storage Conditions

Store at temperature below 30°C. Protect from light and moisture.
Discard 1 month after opening.

Shelf Life:

3 years

Product Description & Packing(s):

A clear and colourless solution available in a LDPE plastic bottle of 5ml.



Manufacturer and Product Registration Holder:
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